No.

Date

Liner Algebra with Applications

第一次作业: P26:T9. T14, P27:T32

9. $\begin{bmatrix} 1 & -1 & 0 & 0 & -4 \\ 0 & 1 & -3 & 0 & -7 \\ 0 & 0 & 1 & -3 & -1 \\ 0 & 0 & 0 & 2 & 4 \end{bmatrix} \xrightarrow{r_4 \times \frac{1}{2}} \begin{bmatrix} 1 & -1 & 0 & 0 & -4 \\ 0 & 1 & -3 & 0 & -7 \\ 0 & 0 & 1 & -3 & -1 \\ 0 & 0 & 0 & 1 & 2 \end{bmatrix} \xrightarrow{r_3+3\times r_4 \to r_3} \begin{bmatrix} 1 & -1 & 0 & 0 & -4 \\ 0 & 1 & -3 & 0 & -7 \\ 0 & 0 & 1 & 0 & 5 \\ 0 & 0 & 0 & 1 & 2 \end{bmatrix}$

$\xrightarrow{r_2+r_3\times 3 \to r_2} \begin{bmatrix} 1 & -1 & 0 & 0 & -4 \\ 0 & 1 & 0 & 0 & 8 \\ 0 & 0 & 1 & 0 & 5 \\ 0 & 0 & 0 & 1 & 2 \end{bmatrix} \xrightarrow{r_1+r_2 \to r_1} \begin{bmatrix} 1 & 0 & 0 & 0 & 4 \\ 0 & 1 & 0 & 0 & 8 \\ 0 & 0 & 1 & 0 & 5 \\ 0 & 0 & 0 & 1 & 2 \end{bmatrix}$

solution $(4, 8, 5, 2)$

14. $\begin{bmatrix} 1 & -3 & 0 & 5 \\ -1 & 1 & 5 & 2 \\ 0 & 1 & 1 & 0 \end{bmatrix} \xrightarrow{r_1+r_2 \to r_2} \begin{bmatrix} 1 & -3 & 0 & 5 \\ 0 & -2 & 5 & 7 \\ 0 & 1 & 1 & 0 \end{bmatrix} \xrightarrow{r_2 \times (-\frac{1}{2})} \begin{bmatrix} 1 & -3 & 0 & 5 \\ 0 & 1 & -\frac{5}{2} & -\frac{7}{2} \\ 0 & 1 & 1 & 0 \end{bmatrix}$

$\xrightarrow{r_3-r_2 \to r_3} \begin{bmatrix} 1 & -3 & 0 & 5 \\ 0 & 1 & -\frac{5}{2} & -\frac{7}{2} \\ 0 & 0 & \frac{7}{2} & \frac{7}{2} \end{bmatrix} \xrightarrow{r_3 \times \frac{2}{7}} \begin{bmatrix} 1 & -3 & 0 & 5 \\ 0 & 1 & -\frac{5}{2} & -\frac{7}{2} \\ 0 & 0 & 1 & 1 \end{bmatrix} \xrightarrow{r_2+\frac{5}{2}r_3 \to r_2} \begin{bmatrix} 1 & -3 & 0 & 5 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 1 \end{bmatrix}$

$\xrightarrow{r_1+3r_2 \to r_1} \begin{bmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 1 \end{bmatrix}$

solution: $(2, -1, 1)$

32. $\begin{bmatrix} 1 & 2 & -5 & 0 \\ 0 & 1 & -3 & -2 \\ 0 & -3 & 9 & 5 \end{bmatrix} \xrightarrow{r_3+3r_2 \to r_3} \begin{bmatrix} 1 & 2 & -5 & 0 \\ 0 & 1 & -3 & -2 \\ 0 & 0 & 0 & -1 \end{bmatrix}$

$\begin{bmatrix} 1 & 2 & -5 & 0 \\ 0 & 1 & -3 & -2 \\ 0 & 0 & 0 & -1 \end{bmatrix} \xrightarrow{r_3 \times (-3) r_2 \to r_3} \begin{bmatrix} 1 & 2 & -5 & 0 \\ 0 & 1 & -3 & -2 \\ 0 & -3 & 9 & 5 \end{bmatrix}$

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第二次作业：P38：3、4、21、22；P48：10、13、23、24、32

3、$\begin{bmatrix}1&2&3&4\\4&5&6&7\\6&7&8&9\end{bmatrix} \xrightarrow{r_2-4r_1\to r_2} \begin{bmatrix}1&2&3&4\\0&-3&-6&-9\\6&7&8&9\end{bmatrix} \xrightarrow{r_3-6r_1\to r_3} \begin{bmatrix}1&2&3&4\\0&-3&-6&-9\\0&-5&-10&-15\end{bmatrix} \xrightarrow[r_2\times(-\frac{1}{3})]{r_2\times(-\frac{1}{3})}$

$\begin{bmatrix}1&2&3&4\\0&1&2&3\\0&1&2&3\end{bmatrix} \xrightarrow{r_3-r_2\to r_3} \begin{bmatrix}1&2&3&4\\0&1&2&3\\0&0&0&0\end{bmatrix}$

Answer: $\begin{bmatrix}①&2&3&4\\0&①&2&3\\0&0&0&0\end{bmatrix}, \begin{bmatrix}①&2&3&4\\4&⑤&6&7\\6&7&8&9\end{bmatrix}$ Pivot columns: 1, 2.

4. $\begin{bmatrix}1&3&5&7\\3&5&7&9\\5&7&9&1\end{bmatrix} \xrightarrow{r_2-3r_1\to r_2} \begin{bmatrix}1&3&5&7\\0&-4&-8&-12\\5&7&9&1\end{bmatrix} \xrightarrow{r_3-3r_1\to r_3} \begin{bmatrix}1&3&5&7\\0&-4&-8&-12\\0&-8&-16&-34\end{bmatrix}$

$\xrightarrow{r_3-2r_2\to r_3} \begin{bmatrix}1&3&5&7\\0&-4&-8&-12\\0&0&0&-10\end{bmatrix} \xrightarrow[r_3\times(-\frac{1}{10})]{r_2\times(-\frac{1}{4})} \begin{bmatrix}1&3&5&7\\0&1&2&3\\0&0&0&1\end{bmatrix}$

Answer: $\begin{bmatrix}①&3&5&7\\0&①&2&3\\0&0&0&①\end{bmatrix}, \begin{bmatrix}①&3&5&7\\3&⑤&7&9\\5&7&9&①\end{bmatrix}$, Pivot columns: 1, 2, 4

21. a. False, Can't.

b. false. This is called Gaussian elimination.

c. True.

d. True.

e. False. The variable x_4 can be 0.

22. a. False. Only the reduced row echelon form is unique.

b. False. They are determined by the positions of leading entries.

c True.

d. False. May be the system is inconsistent.

e. True.

(续)第二次作业

10. $\begin{bmatrix} 4 & 1 & 3 & 9 \\ 1 & -7 & -2 & 2 \\ 8 & 6 & -5 & 15 \end{bmatrix}$ $x_1\begin{bmatrix} 4 \\ 1 \\ 8 \end{bmatrix} + x_2\begin{bmatrix} 1 \\ -7 \\ 6 \end{bmatrix} + x_3\begin{bmatrix} 3 \\ -2 \\ -5 \end{bmatrix} = \begin{bmatrix} 9 \\ 2 \\ 15 \end{bmatrix}$

13. $\begin{bmatrix} 1 & -4 & 2 & 3 \\ 0 & 3 & 5 & -7 \\ -2 & 8 & -4 & -3 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & -4 & 2 & 3 \\ 0 & 3 & 5 & -7 \\ 0 & 0 & 0 & 3 \end{bmatrix}$

The system is inconsistent, so b isn't the combination of the columns.

23a. False. $[-4,3]$ is row vector, $\begin{bmatrix} -4 \\ 3 \end{bmatrix}$ and $(-4,3)$ is column vector

b False. $[-2,5]$ and $[2,-5]$ lie on a line through the origin

c. True.

d. True.

e. False. when $u/v=k$ or v is zero vector, the span$\{u,v\}$ is not a plane,

24. a. True

b. True. $\vec{v}+(\vec{u}-\vec{v})=\vec{u}$

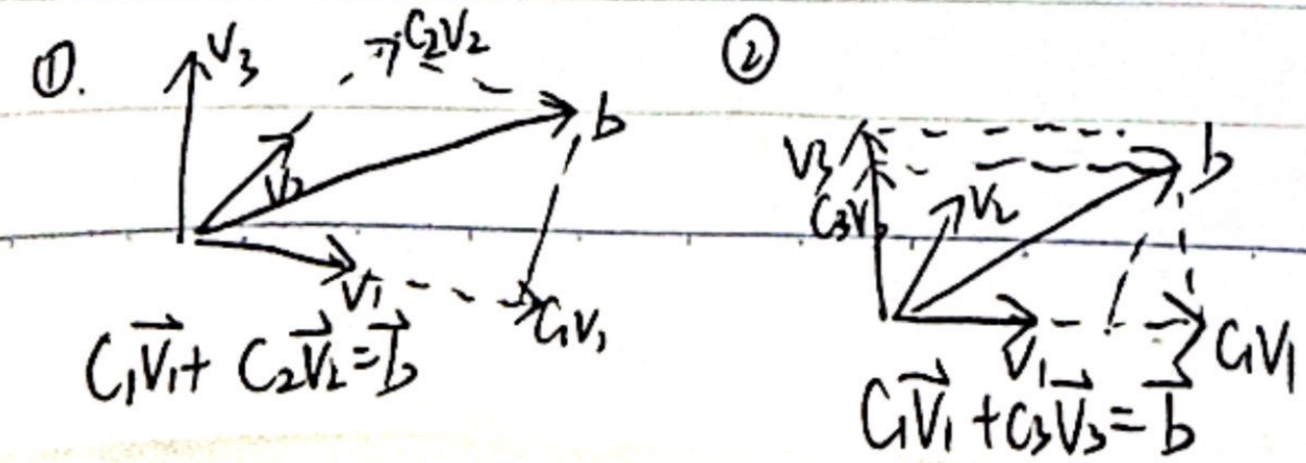
c. False. It can.

d. True.

e. True.

32. The equation $x_1\vec{v}_1+x_2\vec{v}_2+x_3\vec{v}_3=\vec{b}$ have many solutions

The solution is not unique.



KOKUYO